Nithesh Chandher Karthikeyan

Research Engineer, Wallenberg Research Arena for Media and Language

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I aim to specialize and excel in the field of Artificial Intelligence.

WORK EXPERIENCE

Research Engineer
WARA Media and Language, Umeå University

07/2021 - Present — *Umeå, Sweden*

Achievements/Tasks

- Designing, implementing, and maintaining the machine learning platform for WARA Media and Language (WARA ML) arena under the guidance and supervision of Prof. Johanna Björklund and the WARA ML operational team.
- Developing the machine learning platform on top of the OpenStack cloud infrastructure hosted by WARA Common, a sister research arena within Wallenberg AI, Autonomous Systems and Software Program (WASP).
- Collecting and provisioning of data, communication with partners from both academy and industry regarding the development of platform and arena requirements, benchmarking, and facilitating users.
- Assisting the annotation service for the SLAM dataset from Ericsson and with the benchmarking of GENEA 2022 (Generation and Evaluation of Non-verbal Behaviour for Embodied Agents) challenge in the field of animation and motion generation with the main focus on speech-driven gesture synthesis.
- Benchmarking the Visual Object Tracking challenge (VOT 2022) in the field of short/long term visual trackers by providing assistance in evaluating the competition trackers and display a leaderboard showing the competition results.

Guide : Prof. Johanna Björklund - johanna@cs.umu.se

Master Thesis & Research Intern
Uppsala University InfoLab

09/2020 - 06/2021 — *Uppsala, Sweden*

Research Laboratory

Achievements/Tasks

- Carried out Master thesis project in Analysis of visual political communication on YouTube, as a part of the project on Machine Learning and political communication.
- Developed deep learning methods using transfer learning for the analysis of image-based online communication, under the supervision of Prof. Matteo Magnani, the Director of Uppsala University InfoLab, and Prof. Alexandra Segerberg from the Department of Government.
- Developed a custom image classifier to detect object based labels related to climate change using transfer learning.
- Performed information extraction from video frames using the state of the art models like Google's Vision API, Tesseract-OCR, and IBM Watson
- Established a benchmark model by comparing the results with the state-of-the-art models.
- Automated classification of video frames taken from YouTube videos about climate change using the benchmark model.

Guide : Prof. Matteo Magnani, Prof. Alexandra Segerberg

Graduate Teaching Assistant
Uppsala University

09/2020 - 03/2021 — *Uppsala, Sweden*

Achievements/Tasks

- Supported IT and Engineering students for the Data Mining course during Autumn 2020 and Spring 2021.
- Provided assistance to students in executing different Data mining concepts (Data preprocessing, Classification, Clustering, and Association Rule Mining) using the RapidMiner tool.
- Monitored project teams and helped them with their progress.
- Reported a short summary to the professor regarding the major obstacles faced by the students during the project.
- Reviewed project reports and provided suggestions.

Guide : Prof. Matteo Magnani - matteo.magnani@it.uu.se

Machine Learning Intern
Screel Labs

10/2020 - 12/2020

Software Development Company

Achievements/Tasks

- Developed a method to extract information from Hand Written Renal profile using Google Vision API. Image segmentation techniques were used to partition the renal profile table into different segments and Vision API was used to detect the handwritten text in those segments.
- Proposed a solution to assist doctors to convert renal profiles to digital records using Image recognition methods.

EDUCATION

Master of Science in Computer Science
Uppsala University

08/2019 - 06/2021 — *Uppsala, Sweden*

Courses
- Natural Computation for Machine Learning
- Statistical Machine Learning
- Artificial Intelligence
- Data Mining
- Large Datasets for Scientific Application
- Intelligent Interactive Systems

Bachelor of Technology in Computer Science and Engineering
Vellore Institute of Technology

07/2015 - 05/2019 — *Vellore, India, CGPA: 8.19/10*

Courses
- Artificial Intelligence
- Machine Learning
- Internet of Things
- Calculus
- Statistics

TECHNICAL SKILLS

Language	Python, C, C++, HTML, CSS, SQL, Java, Kotlin, R	**Machine Learning**	Predictive Analysis, Statistical Modeling, Deep Learning, Federated Learning, Transfer Learning, Feature Engineering, Classification, Regression
Tools/Packages	NumPy, Pandas, Sklearn, OpenCV, TensorFlow, Keras, Pytorch	**Software/Academic**	Project Management, Scientific Writing

ADDITIONAL SKILLS

OpenStack	Jupyter Notebook	RapidMiner Studio	Weka
Orange	IBM SPSS Software	IBM Watson Studio	Android Studio
IntelliJ	VisionAPI	Tableau	

PROJECTS

Decentralized Air Quality Monitoring and Prediction
- 30 credits project course from Ericsson.
- Developed a solution to predict the concentration of air pollutants (especially PM10) in the city of Stockholm using the Horizontal Federated Machine Learning approach.
- Established the benchmark model by comparing the performance of Federated Learning model and the Centralized model.
- Developed an Android application to display the prediction results and also the work were dockerized to reproduce the results of the model.

Hand Gesture Recognition using Furhat Robot
- Detecting different hand gestures using Deep Learning and create responsive behavior from Furhat Robot for each hand gesture.
- Developed a model to detect the annotations in the videos and predict the gestures based on the annotations using DNN.
- An end-to-end interaction was established using FurhatSDK.

Driver Drowsiness Detection using Deep Learning
- Developed a CNN model which monitors whether the eyes of the user are open or close.
- Designed an alert system to notify the user when his/her eyes are closed for a prolonged period.

Song Classifier using Spotify playlist data
- The mini project aims to classify 200 songs based on whether the person likes them or not with 750 training samples and 13 high level features.
- Established the best performing model by evaluating the performance of different statistical and machine learning models (Logistic Regression, LDA, QDA, k-NN, Boosting and Neural Networks).

RFID Based Library Book Locating and Management System
- Designed a RFID based library book locating and management system using a raspberry pi micro- controller.

Weather Station
- The goal of this project is to build a weather station with real time notifications for climate monitoring, sync it with a cloud platform and analyse weather parameters using Raspberry Pi, Sense HAT, and Google Firebase.

PUBLICATIONS

International Journal of Recent Technology and Engineering (IJRTE)
Intelligent Framework for Public Transport Bus Services system
Author(s)
Nithesh Chandher Karthikeyan, Aswath Ananthasamy, Yokesh Babu Sundaresan
Volume-8 Issue-2S4, July 2019

PERSONAL COURSES

Introduction to Data Science Specialization
Coursera - IBM
- Credential ID: 2XTRKNJZAL77

Databases and SQL for Data Science
Coursera - IBM
- Credential ID: ZVSKH6BZPVRN

Python for Data Science and AI
Coursera - IBM
- Credential ID: D7B6G2H52GT4

Machine Learning Foundations: A Case Study Approach
Coursera - University of Washington
- Credential ID: 8XTL4XBQUTXL

Neural Networks and Deep Learning
Coursera - Deeplearning.ai
- Credential ID: KFUZNJNFWBKM

Convolutional Neural Networks
Coursera- Deeplearning.ai
- Credential ID: E22VR5FMEKHB

EXTRA CURRICULUM

Certificate of Social Service
Certificate of social service from HelpAge India for creating awarness and assisting in raising funds for the care of the elderly people

Certificate of Training in Life Skills
Completed Life Skills Training Programme conducted by Sadhana HR Solutions

Hack4cause
Bronze Echelon certificate for participating in Hack4Cause conducted by IEEE -SSIT (Students Chapter) in VIT University

First level of Active Mechanics
Completed a six months hobby course on robotics conducted by Silicon World

VOLUNTEER EXPERIENCE

Student Coordinator
GraVITas 2017

07/2017 - 09/2017 *Vellore, India*
Registration and Reception Committee

Tasks/Achievements
- Student coordinator of Registration and Reception committee for GraVITas 2017- an International Science Carnival held at Vellore Institute of Technology
- As a student coordinator, I was responsible for making the students to register for the events and to accommodate students coming from other Universities for GraVITas

LANGUAGES

English
Professional Working Proficiency

Swedish
Elementary Proficiency

Tamil
Native or Bilingual Proficiency

REFERENCES

Prof. Johanna Björklund, Associate Professor at Department of Computing Science, Umeå University
Guide : *johanna@cs.umu.se*

Prof. Matteo Magnani, Associate Professor in Computer Science at Uppsala University, Director of Uppsala University InfoLab
Guide : *matteo.magnani@it.uu.se*